

Assignment 3: Data Exploration

Esther Yang

Fall 2025

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (`ECOTOX_Neonicotinoids_Insects_raw.csv`) and the Niwot Ridge NEON dataset for litter and woody debris (`NEON_NIWO_Litter_massdata_2018-08_raw.csv`). Name these datasets “Neonics” and “Litter”, respectively. Be sure to include the sub-command to read strings in as factors.

```
# Set up the R session

# Load necessary packages
library(tidyverse)
library(lubridate)
library(here)

# Check current working directory
here::here()
```

```
## [1] "/home/guest/R/EDE_Fall2025"
```

```
# Read datasets with strings as factors
Neonics <- read.csv(
  here("Data", "Raw", "ECOTOX_Neonicotinoids_Insects_raw.csv"),
  stringsAsFactors = TRUE
)

Litter <- read.csv(
  here("Data", "Raw", "NEON_NIWO_Litter_massdata_2018-08_raw.csv"),
  stringsAsFactors = TRUE
)

Neonics <- Neonics %>% mutate(across(where(is.character), as.factor))
Litter <- Litter %>% mutate(across(where(is.character), as.factor))
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: Neonicotinoids are widely used insecticides that have been linked to adverse effects on non-target insects, including pollinators like bees. Understanding their ecotoxicology helps us assess essential environmental risks and inform regulatory decisions to protect insect biodiversity and ecosystem services.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: Litter and woody debris play crucial roles in forest ecosystems by influencing nutrient cycling, soil formation, carbon storage, and habitat structure. Studying them helps understand ecosystem productivity, decomposition processes, and impacts of disturbances like climate change.

4. How is litter and woody debris sampled as part of the NEON network? Read the [NEON_Litterfall_UserGuide.pdf](#) document to learn more. List three pieces of salient information about the sampling methods here:

Answer: 1. Sampling Equipment and Material Definitions - Litter: Defined as material dropped from the forest canopy with butt end diameter <2cm and length <50cm, collected in elevated 0.5m² PVC traps; Fine woody debris: Defined as material with butt end diameter <2cm and length >50cm, collected in ground traps; Elevated PVC litter trap design follows Smithsonian Tropical Research Institute Center for Tropical Forest Science standards. 2. Spatial Sampling Design - Sampling occurs at terrestrial NEON sites with woody vegetation >2m tall; One litter trap pair (one elevated trap + one ground trap) is deployed per 400m² plot area, resulting in 1-4 trap pairs per plot; Trap placement is either randomized (in sites with >50% woody vegetation cover) or targeted beneath qualifying vegetation (in sites with <50% cover or patchy vegetation). 3. Temporal Sampling Frequency - (a) Ground traps: Sampled once per year; (b) Elevated traps:

Sampling frequency varies by vegetation type: Deciduous forest sites: every 2 weeks during senescence; Evergreen sites: every 1-2 months year-round; In deciduous sites with limited winter access, elevated trap sampling may be discontinued for up to 6 months during dormant season.

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
#getting dimensions of dataset
dim(Neonics)
```

```
## [1] 4623 30
```

6. Using the `summary` function on the “Effect” column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
# determine the most common effects that are studied
# Summary of Effect column, sorted by frequency
sort(summary(Neonics$Effect), decreasing = TRUE)
```

```
##      Population      Mortality      Behavior Feeding behavior
##      1803          1493          360          255
##      Reproduction      Development      Avoidance      Genetics
##      197            136            102            82
##      Enzyme(s)         Growth      Morphology      Immunological
##      62              38            22             16
##      Accumulation      Intoxication      Biochemistry      Cell(s)
##      12              12            11             9
##      Physiology      Histology      Hormone(s)
##      7              5              1
```

Answer: The most common effects (e.g., mortality, growth, reproduction) are fundamental end-points in toxicology as they directly impact population viability and are standardized for regulatory risk assessments.

7. Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed. [TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
#determine the six most commonly studied species in the dataset
summary(Neonics$Species.Common.Name, maxsum = 20)
```

```
##      Honey Bee      Parasitic Wasp
##      667          285
##      Buff Tailed Bumblebee      Carniolan Honey Bee
##      183          152
##      Bumble Bee      Italian Honeybee
```

```
##           140           113
##      Japanese Beetle      Asian Lady Beetle
##           94           76
##      Euonymus Scale      Wireworm
##           75           69
##      European Dark Bee      Minute Pirate Bug
##           66           62
##      Asian Citrus Psyllid      Parastic Wasp
##           60           58
##      Colorado Potato Beetle      Parasitoid Wasp
##           57           51
##      Erythrina Gall Wasp      Beetle Order
##           49           47
## Snout Beetle Family, Weevil      (Other)
##           47           2272
```

```
species_counts <- table(Neonics$Species.Common.Name)
species_counts <- species_counts[names(species_counts) != "(Other)"]

# Get top 6 only
top6_species <- sort(species_counts, decreasing = TRUE)[1:6]
top6_species
```

```
##
##      Honey Bee      Parasitic Wasp Buff Tailed Bumblebee
##      667           285           183
## Carniolan Honey Bee      Bumble Bee      Italian Honeybee
##      152           140           113
```

Answer: These species (e.g., honey bees, Parasitic Wasp) are ecologically important, economically valuable, or serve as model organisms in toxicology studies.

8. Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
# Getting class name of column
class(Neonics$Conc.1..Author.)
```

```
## [1] "factor"
```

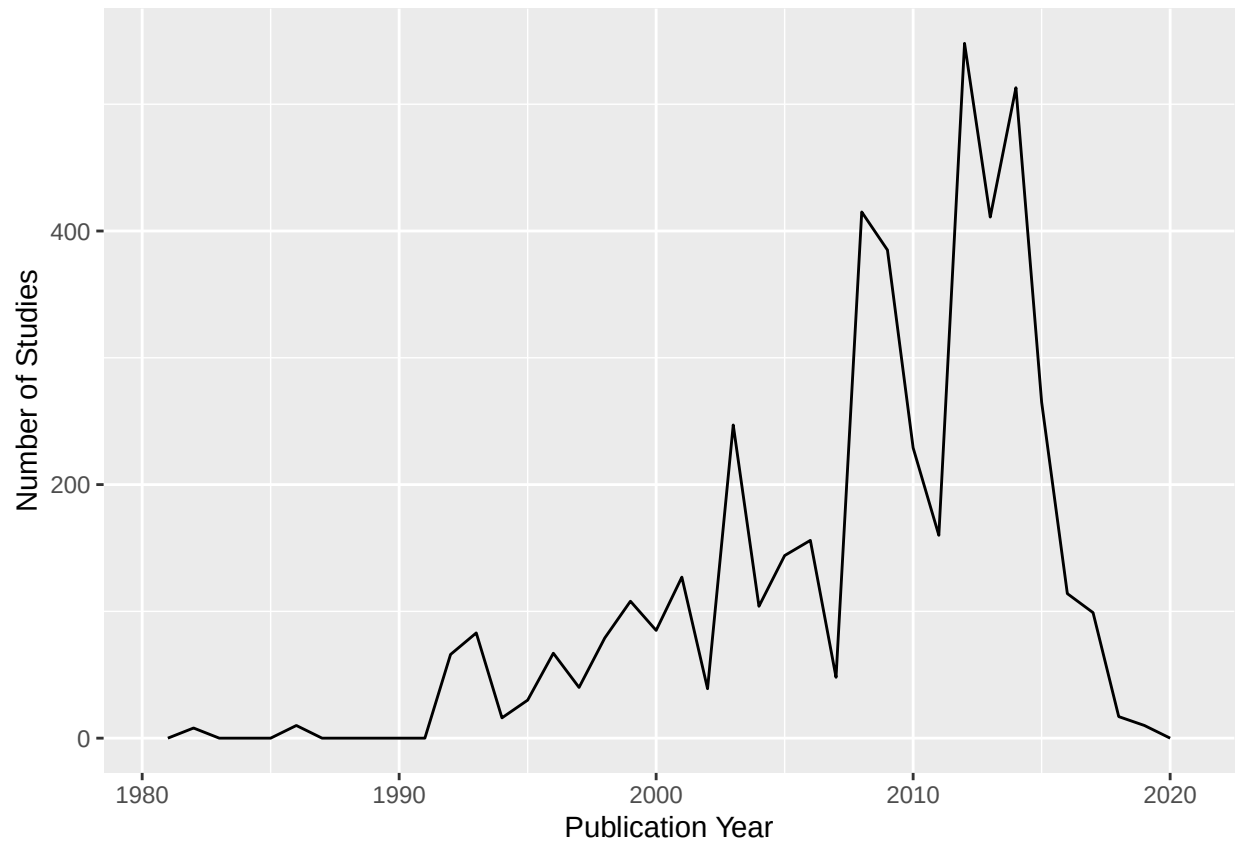
Answer: The concentration column contains non-numeric characters (e.g., “>”, “<”, text descriptors) preventing automatic conversion to numeric.

Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

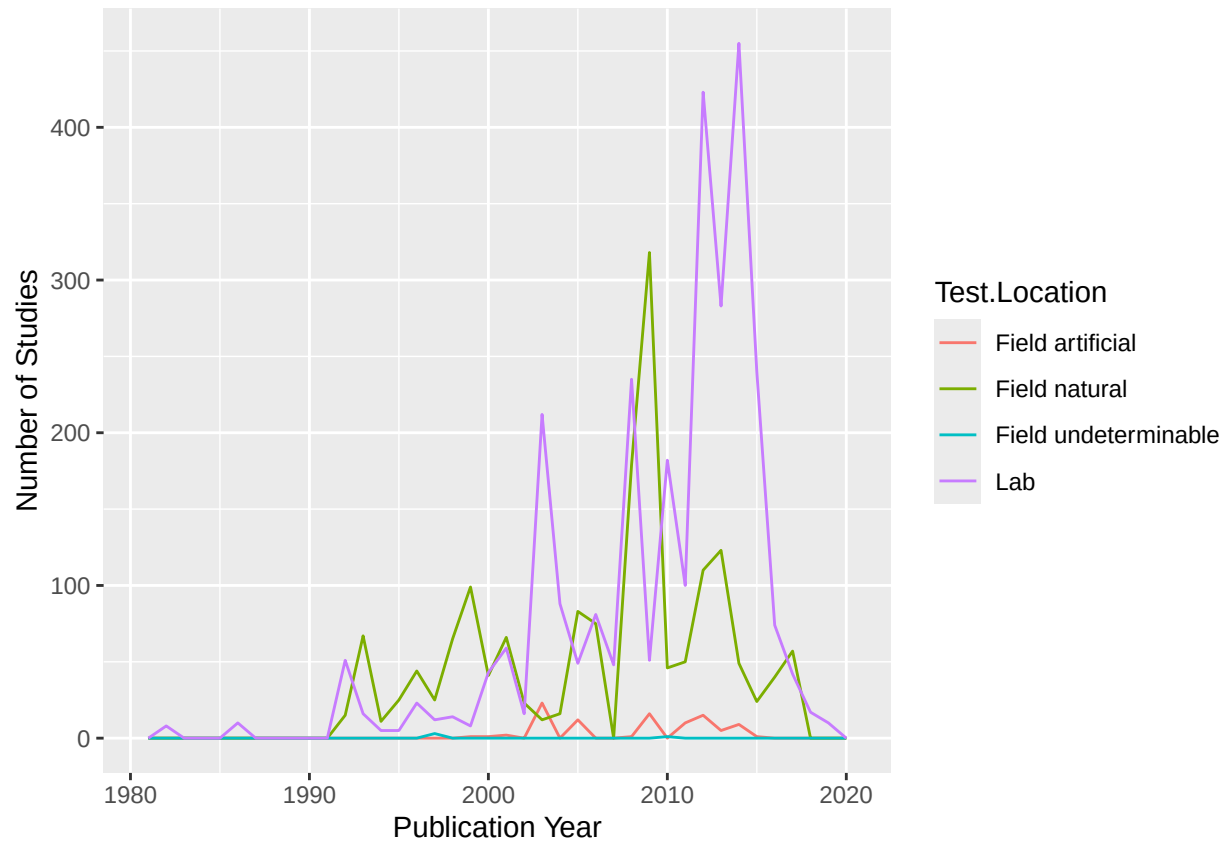
```
# Explore my data graphically (Neonics)

# generate a plot
ggplot(Neonics, aes(x = Publication.Year)) +
  geom_freqpoly(binwidth = 1) +
  labs(x = "Publication Year", y = "Number of Studies")
```



10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
# add a color aesthetic to display different locations with different colors
ggplot(Neonics, aes(x = Publication.Year, color = Test.Location)) +
  geom_freqpoly(binwidth = 1) +
  labs(x = "Publication Year", y = "Number of Studies")
```



Interpret this graph. What are the most common test locations, and do they differ over time?

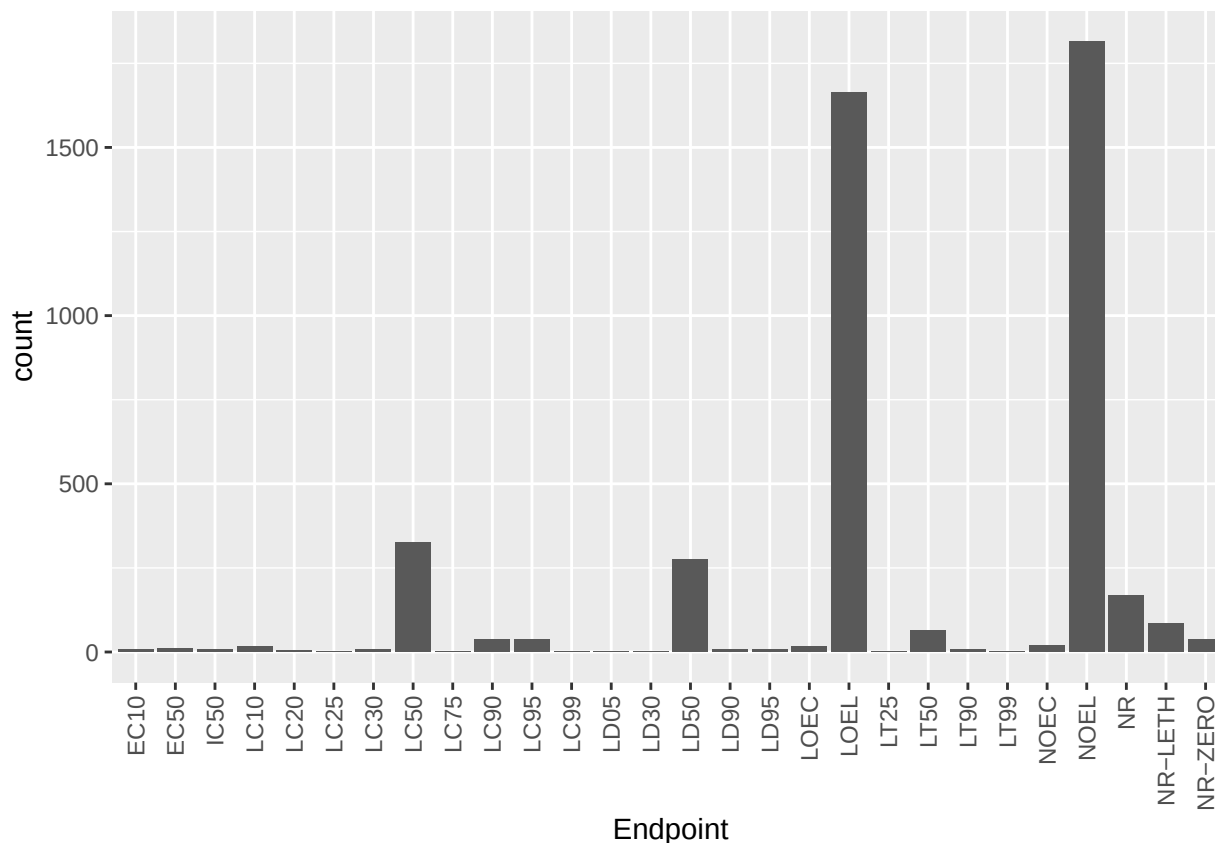
Answer: Laboratory studies are the most common one, with field studies becoming more frequent in recent decades. Test locations have remained relatively consistent over time that lab studies have mostly been the most common one.

11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[**TIP:** Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
# Create a bar graph of Endpoint counts

ggplot(Neonics, aes(x = Endpoint)) +
  geom_bar() +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```



Answer: The two most common end points are LOEL and NOEL. LOEL is defined as Lowest-observable-effect-level: lowest dose (concentration) producing effects that were significantly different. NOEL is defined as No Observed Effects Residue: The highest residue concentration producing effects not significantly different from responses of controls according to author's reported statistical test.

Explore your data (Litter)

- Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
# Explore my data (Litter)
```

```
# Determine the class of collectDate
class(Litter$collectDate)
```

```
## [1] "factor"
```

```
# Convert to Date
Litter$collectDate <- as.Date(Litter$collectDate)
class(Litter$collectDate)
```

```
## [1] "Date"
```

```
# determine which dates litter was sampled in August 2018
unique(Litter$collectDate)
```

```
## [1] "2018-08-02" "2018-08-30"
```

13. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

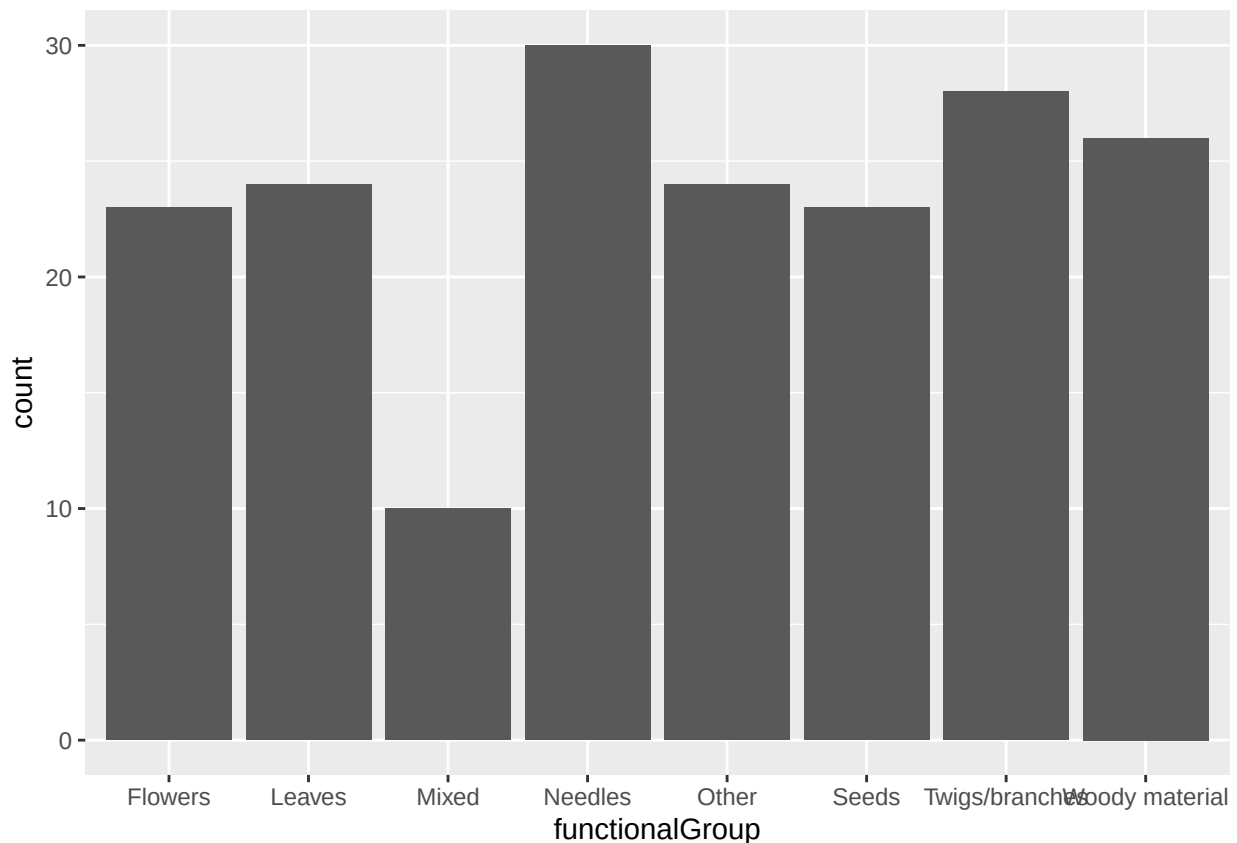
```
#list the different plotIDs sampled at Niwot Ridge
unique(Litter$plotID)
```

```
## [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051
## [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

Answer: `unique` lists all distinct values, while `summary` would show frequency counts and might truncate factor levels.

14. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

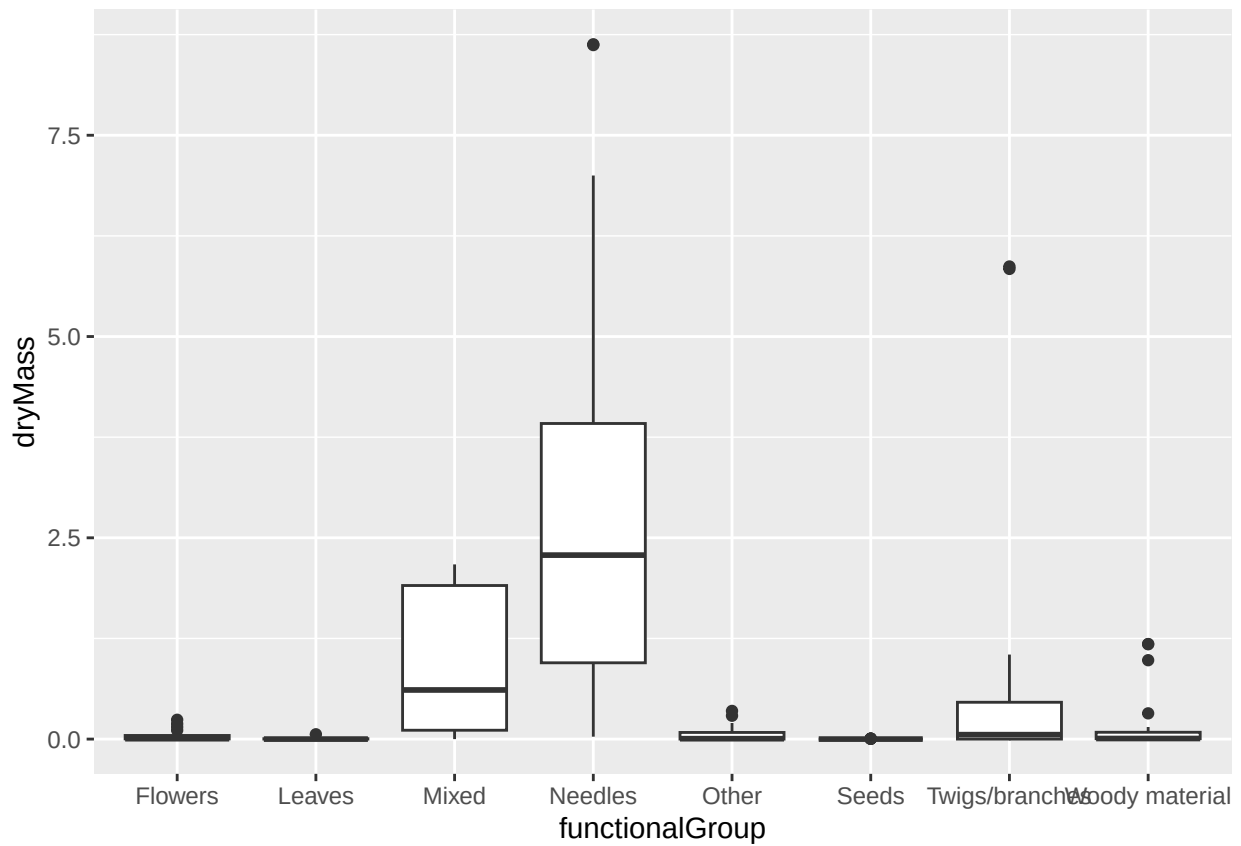
```
# Create a bar graph of functionalGroup counts
ggplot(Litter, aes(x = functionalGroup)) +
  geom_bar()
```



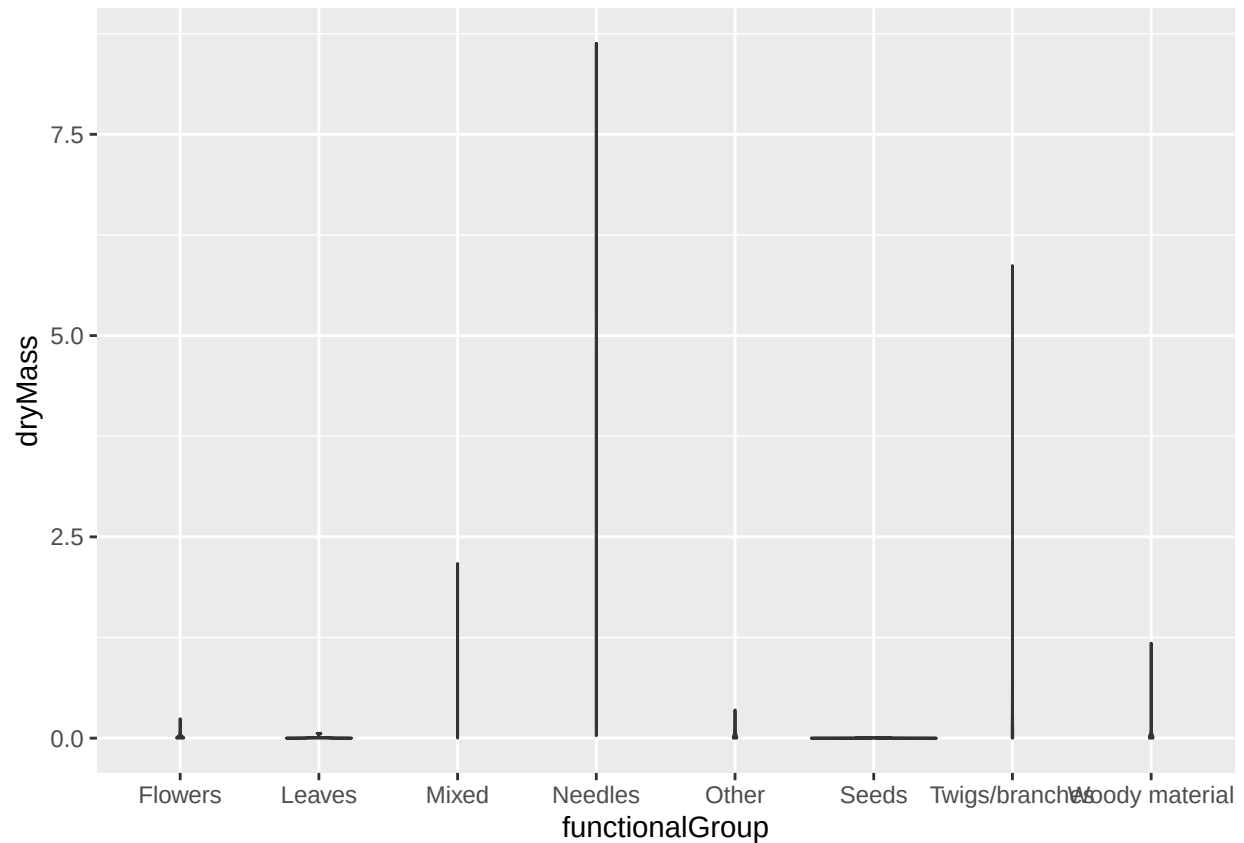
15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by `functionalGroup`.

```
# create a boxplot and a violin plot of dryMass by functionalGroup
```

```
# Boxplot  
ggplot(Litter, aes(x = functionalGroup, y = dryMass)) +  
  geom_boxplot()
```



```
# Violin plot  
ggplot(Litter, aes(x = functionalGroup, y = dryMass)) +  
  geom_violin()
```



Why is the boxplot a more effective visualization option than the violin plot in this case?

Answer: The boxplot is more effective with median, quartiles, and outliers. The violin plot obscures details due to the high number of zero values and extreme outliers.

What type(s) of litter tend to have the highest biomass at these sites?

Answer: Needles tend to have the highest biomass at Niwot Ridge sites, followed by the mixed functional group.